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SECTION:	S-8
SUBJECT:	FULL STACK APPLICATION DEVELOPMENT

#SKILL - 10 React State Management using useState Object

Tasks:

1. **Create a new React component named StudentManager.**

   <u>Output:-</u>

```
PS C:\Users\sathv\Downloads\full stack labs\exp10> npx create-react-app student-manager

Creating a new React app in C:\Users\sathv\Downloads\full stack labs\exp10\student-manager.

Installing packages. This might take a couple of minutes.
Installing react, react-dom, and react-scripts with cra-template...
```

```
EXP10
  student-manager
    node_modules
    public
    src
      App.css
      App.js
      App.test.js
      index.css
      index.js
      logo.svg
      reportWebVitals.js
      setupTests.js
      StudentManager.css
      StudentManager.js
    .gitignore
    package-lock.json
    package.json
    README.md
```

2. Inside the component, create an initial students array with at least 5 objects containing:
a. id
b. name
c. course

<u>Output:-</u>

```
# StudentManager.css    JS StudentManager.js ×    JS App.js
student-manager > src > JS StudentManager.js > default
import React, { useState } from "react";
import "./StudentManager.css";

function StudentManager() {

  const [students, setStudents] = useState([
    { id: 1, name: "Sathvik", course: "CSE" },
    { id: 2, name: "Rahul", course: "ECE" },
    { id: 3, name: "Ananya", course: "IT" },
    { id: 4, name: "Kiran", course: "MECH" },
    { id: 5, name: "Priya", course: "CSE" }
  ]);

  const [newStudent, setNewStudent] = useState({
    id: "",
    name: "",
    course: ""
  });

  const handleChange = (e) => {
    setNewStudent({
      ...newStudent,
      [e.target.name]: e.target.value
    });
  };

  const addStudent = () => {
    setStudents([...students, newStudent]);

    setNewStudent({
      id: "",
      name: "",
      course: ""
    });
  };

  const deleteStudent = (id) => {
    const updatedList = students.filter(student => student.id !== id);
    setStudents(updatedList);
  };
```

3. Use useState to store:
a. students (array of student objects)
b. newStudent (object with fields like id, name, course)

Output:-

```
function StudentManager() {

  const [students, setStudents] = useState([
    { id: 1, name: "Sathvik", course: "CSE" },
    { id: 2, name: "Rahul", course: "ECE" },
    { id: 3, name: "Ananya", course: "IT" },
    { id: 4, name: "Kiran", course: "MECH" },
    { id: 5, name: "Priya", course: "CSE" }
  ]);

  const [newStudent, setNewStudent] = useState({
    id: "",
    name: "",
    course: ""
  });
```

4. Create input boxes to accept id, name, and course.

i. Update the newStudent object using onChange events.

Output:-

```
<div className="container">

  <h2>Student Manager</h2>

  <input name="id" placeholder="ID" value={newStudent.id} onChange={handleChange}/>
  <input name="name" placeholder="Name" value={newStudent.name} onChange={handleChange}/>
  <input name="course" placeholder="Course" value={newStudent.course} onChange={handleChange}/>

  <button onClick={addStudent}>Add Student</button>

  {students.length === 0 ? (
    <p>No students available</p>
  ) : (
    <table>

      <thead>
        <tr>
          <th>ID</th>
          <th>Name</th>
          <th>Course</th>
          <th>Action</th>
        </tr>
      </thead>
```

5. Create an Add Student button.

i. On click, add the newStudent object to the students array.

ii. After adding, clear the input fields.

Output:-

```
const addStudent = () => {
  setStudents([...students, newStudent]);

  setNewStudent({
    id: "",
    name: "",
    course: ""
  });
};

const deleteStudent = (id) => {
  const updatedList = students.filter(student => student.id !== id);
  setStudents(updatedList);
};

return (
  <div className="container">

    <h2>Student Manager</h2>

    <input name="id" placeholder="ID" value={newStudent.id} onChange={handleChange}/>
    <input name="name" placeholder="Name" value={newStudent.name} onChange={handleChange}/>
    <input name="course" placeholder="Course" value={newStudent.course} onChange={handleChange}/>

    <button onClick={addStudent}>Add Student</button>

    {students.length === 0 ? (
      <p>No students available</p>
```

6. Display the list of students below using the students state.

Output:-

```
<tbody>
  {students.map((s) => (
    <tr key={s.id}>
      <td>{s.id}</td>
      <td>{s.name}</td>
      <td>{s.course}</td>
      <td>
```

7. Each student row should have a Delete button.
i. On clicking delete, remove the record from the students array.
ii. The UI should update immediately using setStudents.

Output:-

```
<thead>
  <tr>
    <th>ID</th>
    <th>Name</th>
    <th>Course</th>
    <th>Action</th>
  </tr>
</thead>

<tbody>
  {students.map((s) => (
    <tr key={s.id}>
      <td>{s.id}</td>
      <td>{s.name}</td>
      <td>{s.course}</td>
      <td>
        <button onClick={() => deleteStudent(s.id)}>Delete</button>
      </td>
    </tr>
  ))}
</tbody>

</table>

v>
```

8. If the list becomes empty, display the message: "No students available"

Output:-

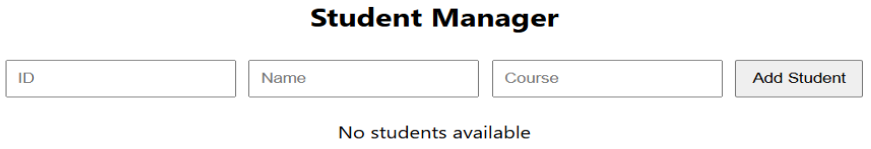
```
return (
  <div className="container">

    <h2>Student Manager</h2>

    <input name="id" placeholder="ID" value={newStudent.id} onChange={handleChange}/>
    <input name="name" placeholder="Name" value={newStudent.name} onChange={handleChange}/>
    <input name="course" placeholder="Course" value={newStudent.course} onChange={handleChange}/>

    <button onClick={addStudent}>Add Student</button>

    {students.length === 0 ? (
      <p>No students available</p>
    ) : (        <table>
```



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9. Add basic CSS styling for a clean UI (margin, padding, table style, buttons).

Output:-

StudentManager.css | StudentManager.js M | index.css | App.js

student-manager > src > StudentManager.css > th

```css
.container{
width:60%;
margin:auto;
text-align:center;
}

input{
margin:5px;
padding:8px;
}

button{
padding:8px 12px;
margin:5px;
}

table{
width:100%;
margin-top:20px;
border-collapse:collapse;
}

th,td{
border:1px solid gray;
padding:10px;
}

th{
background: #eee;
}
```

Student Manager

ID | Name | Course | Add Student

ID	Name	Course	Action
2400032783	sathvik	fsad	Delete
2400032783	jogi	fasd	Delete
2400031075	sai	fsad	Delete
2400032518	satwik	fsad	Delete

GITHUB REPO :- https://github.com/PulluruSathvik/skillab10